Finding Sign and Like-Term Errors in Polynomial Work Answer Key

Algebra 1

Quick Answers

1. $3x + 10$
2. $3x^2 + 4x$
3. (b) value at x = 2 = -2, (c) corrected expression = $3x^2 - 7x$, —
4. $7x^3 + 3x^2$
5. (b) value at x = 2 = 28, (c) corrected expression = $5x^2 + 4x$, —
6. (b) value at x = 2 = 1, (c) corrected expansion = $2x^2 - 11x + 15$, —
7. $2x^2 + 2x + 7$
8. (b) value at x = 3 = 44, (c) corrected expression = $8x + 20$, —

What is verified

4 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 3, 5, 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1
HSA-APR.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

3. *If they got 14: kept +4x instead of subtracting it, so the minus reached only the first term*
5. *If they got 72: added the unlike terms and added the exponents too*
6. *If they got −29: made the product of the two negative constants negative*
8. *If they got 12: dropped the middle term of the square and did not distribute the minus*
8. *If they got 36: squared correctly but did not distribute the minus over the second product*

1. Simplify $(4x + 7) - (x - 3)$.

Solution: The minus sign applies to both terms of the second bracket, so $-(x - 3)$ becomes $-x + 3$. Then $4x - x = 3x$ and $7 + 3 = 10$. $\boxed{3x + 10}$

2. Simplify $5x^2 + 3x - 2x^2 + x$.

Solution: Only terms with the same power may be combined. Squares: $5x^2 - 2x^2 = 3x^2$. Linear: $3x + x = 4x$ (a bare x has coefficient 1). The two groups stay separate. $\boxed{3x^2 + 4x}$

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3. Dev: $(5x^2 - 3x) - (2x^2 + 4x)$ written as $3x^2 + x$. Find the error, test it, correct it.

Solution (a) — instructor-judged. Dev distributed the minus sign to the first term of the second bracket only. He used $-2x^2$ correctly but carried $+4x$ across unchanged; it should be $-4x$. The correct middle step is $5x^2 - 3x - 2x^2 - 4x$. Full credit names the distributing-the-minus error or writes that middle step; "the answer is wrong" does not.

Solution (b): Substituting $x = 2$ into the original: $(5 \cdot 4 - 3 \cdot 2) - (2 \cdot 4 + 4 \cdot 2) = (20 - 6) - (8 + 8) = 14 - 16$. $\boxed{-2}$

Dev's expression gives $3 \cdot 4 + 2 = 14$ at $x = 2$, so the test value exposes the error.

Solution (c): $5x^2 - 3x - 2x^2 - 4x = 3x^2 - 7x$. $\boxed{3x^2 - 7x}$

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4. Simplify $6x^3 - 2x^2 + x^3 + 5x^2$.

Solution: Cubes together and squares together: $6x^3 + x^3 = 7x^3$ and $-2x^2 + 5x^2 = 3x^2$. A cube and a square are never like terms, however close they sit on the page. $\boxed{7x^3 + 3x^2}$

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5. Mara: $3x^2 + 4x + 2x^2$ written as $9x^3$. Find the error, test it, correct it.

Solution (a) — instructor-judged. Two things went wrong in one step. First, $4x$ was treated as a like term with the two squared terms; it has a different power and cannot join them. Second, the exponents were added along with the coefficients — combining like terms adds coefficients only and leaves the power alone. Full credit names both halves, or states the like-term rule that rules the step out.

Solution (b): At $x = 2$: $3 \cdot 4 + 4 \cdot 2 + 2 \cdot 4 = 12 + 8 + 8$. $\boxed{28}$

Mara's $9x^3$ gives $9 \cdot 8 = 72$ at $x = 2$, nowhere near 28.

Solution (c): Only the two squared terms combine: $3x^2 + 2x^2 = 5x^2$, and $4x$ stays as it is. $\boxed{5x^2 + 4x}$

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6. Theo: $(2x - 5)(x - 3)$ expanded as $2x^2 - 11x - 15$. Find the error, test it, correct it.

Solution (a) — instructor-judged. The two constants are both negative, so their product is positive: $(-5)(-3) = +15$. Theo wrote -15. Everything else in his answer is right — the $2x^2$ and the $-11x$ from $-6x - 5x$ both check out. Full credit points at the product of the two negative constants.

Solution (b): At $x = 2$: $(2 \cdot 2 - 5)(2 - 3) = (-1)(-1)$. $\boxed{1}$

Theo's expression gives $8 - 22 - 15 = -29$ at $x = 2$, so one substitution settles it.

Solution (c):

$$(2x - 5)(x - 3) = 2x^2 - 6x - 5x + 15 \;=\; 2x^2 - 11x + 15.$$

$\boxed{2x^2 - 11x + 15}$

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7. Simplify $(3x^2 - 5x + 2) - (x^2 - 5x - 6) + (2x - 1)$.

Solution: Distribute the minus over all three terms of the middle bracket; the last bracket is added, so it is unchanged:

$$\begin{aligned}
&3x^2 - 5x + 2 - x^2 + 5x + 6 + 2x - 1 \\
&= (3x^2 - x^2) + (-5x + 5x + 2x) + (2 + 6 - 1) \\
&= 2x^2 + 2x + 7.
\end{aligned}$$

Note that $-5x$ and $+5x$ cancel, so the surviving linear term comes only from the last bracket.

$\boxed{2x^2 + 2x + 7}$

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8. Student line: $(x+4)^2 - (x-2)(x+2) = x^2 + 16 - x^2 - 4 = 12$. Find both errors, test, correct.

Solution (a) — instructor-judged. Error one: $(x+4)^2$ was written as $x^2 + 16$. A squared binomial always has a middle term, $2 \cdot x \cdot 4 = 8x$, so it is $x^2 + 8x + 16$. Error two: the minus in front of the second product reached only the x^2. Since $(x-2)(x+2) = x^2 - 4$, subtracting it gives $-x^2 + 4$, not $-x^2 - 4$. Full credit names both; naming one earns half credit.

Solution (b): At $x = 3$: $(3+4)^2 - (3-2)(3+2) = 49 - (1)(5) = 49 - 5$. $\boxed{44}$

The student's line gives 12, which is not even a function of x — a warning sign in itself, since the original expression plainly still contains x.

Solution (c):

$$\begin{aligned}
(x+4)^2 - (x-2)(x+2) &= (x^2 + 8x + 16) - (x^2 - 4) \\
&= x^2 + 8x + 16 - x^2 + 4 \;=\; 8x + 20.
\end{aligned}$$

Check at $x = 3$: $8 \cdot 3 + 20 = 44$, matching part (b).

$\boxed{8x + 20}$